Erkan Karabulut

PhD Candidate in Computer Science, University of Amsterdam

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Research Profile

I research knowledge discovery: finding previously unknown, interesting, and potentially useful patterns from various data types, including tables, graphs, and more recently images. Some of the topics I worked on are finding co-occurrence patterns, utilizing background knowledge, and leveraging foundation models. The methods I developed are neuro-symbolic, neural for scalability and symbolic for interpretable knowledge discovery. A recent highlight in my research is the reframing of rule mining as a neuro-symbolic learning problem, shifting the focus from '*how often does a set of features co-occur?*' to '*what does a learned model believe about their conditional dependency?*', bridging data mining and machine learning. In the future, I aim to broaden my research to new problems in artificial intelligence, while carrying the knowledge discovery perspective with me.

Education

2023 – present	**Ph.D. in Computer Science**, University of Amsterdam, The Netherlands Thesis: *Neurosymbolic Knowledge Discovery from Structured Data*
2019 – 2022	**M.Sc. in Computer Science**, Technical University of Munich, Germany Thesis: *Machine Learning-based Data Classification and Data Aggregation on the Edge*
2014 – 2019	**B.Sc. in Computer Engineering**, Yildiz Technical University, Turkey Thesis: *Adaptive Learning-based Tracing Tool for Weather Research and Forecasting Software*

Work Experience

Mar 2026 – Dec 2026	**Visiting Researcher** Learning and Reasoning Group, Vrije Universiteit Amsterdam
Nov 2025 – Jan 2026	**Visiting Researcher** Translational AI Lab, Amsterdam University Medical Center
Feb 2023 – Jan 2027	**PhD Candidate** Intelligent Data Engineering Lab, University of Amsterdam
Oct 2020 – Sep 2022	**Research Assistant** Research Institute of the Free State of Bavaria for Software-Intensive Systems (fortiss), Germany
Nov 2019 – Oct 2020	**Working Student, Software Infrastructure** Planerio GmbH, Germany
Feb 2018 – Sep 2019	**Software Developer (Part-time)** IPera Solutions, Turkey
Sep 2017 – Jul 2018	**Student Research Assistant** Yildiz Technical University, Turkey

Selected Publications

Karabulut, **Erkan**, Daniel Daza, Paul Groth, M.C. Schut, and Victoria Degeler. "Model-Agnostic Association Rule Learning with Tabular Foundation Models." *Transactions on Machine Learning Research*, 2026. openreview.net/forum?id=Y70lnNnA6T.

Karabulut, **Erkan**, Daniel Daza, Emile van Krieken, Paul Groth, and Victoria Degeler. "Differentiable Numerical Association Rule Learning." *Transactions on Machine Learning Research*, 2026, under review. openreview.net/forum?id=0WZf2uidpa.

Karabulut, **Erkan**, Paul Groth, and Victoria Degeler. "Learning semantic association rules from internet of things data." *Neurosymbolic Artificial Intelligence*, vol. 1, 2025. https://doi.org/10.1177/29498732251377518.

Karabulut, **Erkan**, Paul Groth, and Victoria Degeler. "Neurosymbolic Association Rule Mining from Tabular Data." *Proceedings of the 19th International Conference on Neurosymbolic Learning and Reasoning (NeSy)*. Proceedings of Machine Learning Research 284, PMLR, 2025, pp. 565–88, proceedings.mlr.press/v284/karabulut25a.html.

Karabulut, **Erkan**, Salvatore F. Pileggi, Paul Groth, and Victoria Degeler. "Ontologies in digital twins: A systematic literature review." *Future Generation Computer Systems*, vol. 153, 2024, pp. 442–56. https://doi.org/10.1016/j.future.2023.12.013.

Other Publications

Abbes, Jasper, **Erkan Karabulut**, and Victoria Degeler. "Rule Learning on Sensor Data in Cyber-Physical Systems and Digital Twins: A Systematic Review." Under submission at ACM Computing Surveys (CSUR), 2026.

Karabulut, **Erkan**, Daniel Daza, Paul Groth, and Victoria Degeler. "Discovering Association Rules in High-Dimensional Small Tabular Data." *1st International Workshop on Advanced Neuro-Symbolic Applications (ANSyA), at ECAI 2025*. CEUR Workshop Proceedings 4125, CEUR-WS, 2025, pp. 105–13, ceur-ws.org/Vol-4125/paper_26.pdf.

Karabulut, **Erkan**, Paul Groth, and Victoria Degeler. "PyAerial: Scalable association rule mining from tabular data." *SoftwareX*, vol. 31, 2025, p. 102341. https://doi.org/10.1016/j.softx.2025.102341.

Degeler, Victoria, Mostafa Hadadian, **Erkan Karabulut**, Alexander Lazovik, Hester van het Loo, Andrés Tello, and Huy Truong. "DiTEC: Digital twin for evolutionary changes in water distribution networks." *Leveraging Applications of Formal Methods, Verification and Validation. Application Areas (ISoLA 2024)*. Lecture Notes in Computer Science 15223, Springer, 2024, pp. 62–82, https://doi.org/10.1007/978-3-031-75390-9_5.

Huang, Yiwen, **Erkan Karabulut**, and Victoria Degeler. "Large Language Model for Ontology Learning in Drinking Water Distribution Network Domain." *Workshop on Evaluation of Language Models in Knowledge Engineering (ELMKE), November 26–28, 2024. Co-located with EKAW 2024*. CEUR Workshop Proceedings, CEUR-WS.org, 2024, ceur-ws.org/Vol-3967/ELMKE_2024_paper_1.pdf.

Karabulut, **Erkan**, Paul Groth, and Victoria Degeler. "3K: Knowledge-Enriched Digital Twin Framework." *Proceedings of the 14th International Conference on the Internet of Things*. ACM, 2024, pp. 188–93, https://doi.org/10.1145/3703790.3703834.

Karabulut, **Erkan**, Victoria Degeler, and Paul Groth. "Semantic Association Rule Learning from Time Series Data and Knowledge Graphs." *Proceedings of the Second International Workshop on Semantic Industrial Information Modelling (SemIIM), at ISWC 2023*. CEUR Workshop Proceedings 3647, CEUR-WS, 2023, ceur-ws.org/Vol-3647/SemIIM2023_paper_3.pdf.

Karabulut, **Erkan**, and Rute C. Sofia. "An Analysis of Machine Learning-Based Semantic Matchmaking." *IEEE Access*, vol. 11, 2023, pp. 27829–42. https://doi.org/10.1109/ACCESS.2023.3259360.

Bnouhanna, Nisrine, **Erkan Karabulut**, Rute C. Sofia, Erin E. Seder, Gabriele Scivoletto, and Gianluca Insolvibile. "An evaluation of a semantic thing to service matching approach in industrial IoT environments." *2022 IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events (PerCom Workshops)*. IEEE, 2022, pp. 433–38, https://doi.org/10.1109/PerComWorkshops53856.2022.9767519.

Karabulut, **Erkan**, Nisrine Bnouhanna, and Rute C. Sofia. "ML-based data classification and data aggregation on the edge." *Proceedings of the CoNEXT Student Workshop*. ACM, 2021, pp. 21–22, https://doi.org/10.1145/3488658.3493786.

Yazici, Ilkay Melek, **Erkan Karabulut**, and Mehmet S. Aktas. "A data provenance visualization approach." *2018 14th International Conference on Semantics, Knowledge and Grids (SKG)*. IEEE, 2018, pp. 84–91, https://doi.org/10.1109/SKG.2018.00019.

Software

PyAerial	Python library for scalable association rule mining from tabular data github.com/DiTEC-project/pyaerial
wdn-knowledge-graph	Python library for converting water distribution network (WDN) data (EPANET `.inp` files) into RDF knowledge graphs using our WDN ontology github.com/DiTEC-project/wdn-knowledge-graph

Teaching and Supervision

Teaching Assistance

2025	Semantic Web, University of Amsterdam
2024	Information Organization, University of Amsterdam
2024	Knowledge and Data (Semantic Web), University of Amsterdam
2023	Knowledge and Data (Semantic Web), University of Amsterdam

Thesis Supervision

2026, M.Sc.	Jasper Abbes, *To be decided.*
2025, M.Sc.	Yujie Cao. *Rule Learning for Water Distribution Leak Detection*
2025, M.Sc.	Louise Buijs. *Semantically enriched association rule mining as a tool for finding better interpretable rules in gene expression data*
2025, M.Sc.	Jose Pérez Hernández. *Sensor Data Analysis: A Research on Missing Value Imputation with Semantics*
2025, M.Sc.	Yonathan Adye. *Noise Detection in Sensor Data Analysis with Semantics: A Clustering-Based Approach*
2024, M.Sc.	Rhea Huang. *Large Language Model for Ontology Learning in Drinking Water Distribution Network Domain*
2024, B.Sc.	Alessio Silvério. *Heuristic-based Numerical Association Rule Mining*

Talks, Tutorials and Research Dissemination

Dec 2026	*Tutorial on Composition patterns in Digital Twin Representation Models* The 24th International Conference on Service-Oriented Computing
Jun 2026	*Scalable Knowledge Discovery* WAI Seminar at VU Computer Science Department
Jan 2026	*Scalable Knowledge Discovery from Tabular Data* Table Representation Learning (TRL) Seminar at CWI
Dec 2025	*Scalable Neurosymbolic Knowledge Discovery from Tabular Data* VU Machine Learning Seminar

Nov 2025	*Semantic Rule Learning for Knowledge Discovery and Interpretable Inference* Amsterdam Data Science Center Seminar
Sep 2025	*Knowledge Discovery via Neurosymbolic Rule Learning* Unlimidata
Jul 2025	*Semantic Rule Learning for Knowledge Discovery and Interpretable Inference* SICK AG
May 2025	*Semantic Rule Learning for Knowledge Discovery and Interpretable Inference* Amsterdam UMC Translational AI Lab

Academic Service

Program Committee	Amsterdam AI Thesis Award 2023 ICSOC 2025 (Artifact Evaluation) EurIPS 2025 Workshop on AI for Tabular Data
Reviewer	ECAI 2024 CIKM 2023, 2024, 2025, 2026 Journal of Machine Learning Research (JMLR)